



Yashwanth Krishna Gampa

Roll No.:254103107

M.Tech - Mechanical Engineering

B.Tech - Aerospace Engineering

Specializing in Computational Mechanics

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[PORTFOLIO URL](#)

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An engineer with hands-on experience spanning numerical simulation and mechanical design. Passionate about utilizing robust CAD/CAE workflows to solve structural and fluid-dynamic challenges in innovative Design and Analysis roles.

EDUCATION

Degree/Certificate	Institute/Board	CGPA/Percentage	Year
M.Tech	Indian Institute of Technology, Guwahati	9.04 (Current)	2025 – Present
B.Tech	Lovely Professional University, Phagwara	8.85	2018 – 2022
Senior Secondary	Board of Intermediate Education Telangana	93.0%	2018
Secondary	Board of Secondary Education Telangana	8.70	2016

EXPERIENCE

- Asteria Aerospace Limited** Jun. 2022 – Jun. 2023
Associate Engineer I & Intern: Flight Operations Bengaluru
 - Executed large-scale enterprise aerial surveys covering 2,200+ km of pipeline networks (GAIL, Pipeline Infra Ltd.).
 - Engineered VTOL flight paths from KML data, processed PPK payload data for high-resolution GIS orthomosaic generation.
- Mechathon Engineering** May 2021 – Jul. 2021
Design Intern Kanchipuram
 - Engineered and assembled a 40-component high-pressure plunger pump in Creo with precise spatial alignment.
 - Generated comprehensive 2D manufacturing drafts with strict GD&T for all parts to facilitate production.

PROJECTS

- High-Performance Parallelization of an Unstructured NSF Solver | [Fortran, OpenACC]** Jun. 2026 – Present
Prof. T. K. Mankodi / M.Tech Project [View Project](#)
 - Executing GPU porting of a legacy cell-centered Finite Volume Fortran solver using OpenACC for Navier-Stokes-Fourier equations on unstructured meshes.
 - Optimizing parallel performance with the objective of drastically reducing simulation turnaround times to enable scalable, high-fidelity thermal and aerodynamic analysis for complex industrial geometries.
- 2D Incompressible Navier-Stokes Solver | [Fortran, OpenMP, OpenACC, ANSYS]** Jun. 2026 – Aug. 2026
Personal Project [View Project](#)
 - Engineered a 2D fractional-step CFD solver on a staggered grid for cavity and cylinder flows, validating drag coefficients against ANSYS Fluent.
 - Accelerated computations via OpenMP and OpenACC, achieving 3.2x and 4.05x respective speedups over serial execution.
- 2D Heat Conduction Solvers & Parallelization | [C, MPI, OpenMP]** Apr. 2026 – May 2026
Prof. A. K. Soti / Advanced Computational Fluid Dynamics [View Project](#)
 - Discretized and solved the 2D heat conduction equation on finite difference grids, engineering Conjugate Gradient (CG) and Preconditioned CG (PCG) solvers that accelerated convergence by 85-90%.
 - Parallelized Jacobi and Red-Black Gauss-Seidel solvers via MPI and OpenMP, achieving up to 3.23x CPU speedup.
- Turbine Blade Cooling Analysis Using Liquid Coolants | [CATIA, ANSYS]** Jan. 2022 – May 2022
Lovely Professional University [View Project](#)
 - Modified a turbine blade model to integrate 18 internal liquid cooling passages using CATIA V5.
 - Conducted thermal analysis in ANSYS to benchmark heat transfer performance of liquid coolants versus air-cooled designs.

TECHNICAL SKILLS

- Programming:** C, Fortran, Python (Intermediate)
- Analytical & Computational:** Computational Fluid Dynamics (CFD), Numerical Methods
- CAD Tools:** PTC Creo, CATIA V5, SolidWorks (Elementary)
- CAE Tools:** ANSYS Workbench (Static Structural, Fluent) (Elementary)
- Parallel Computing:** MPI, OpenMP, OpenACC (Intermediate)

KEY COURSES TAKEN

- Computational Fluid Dynamics (CFD)
- Numerical Analysis
- Finite Element Method (FEM)
- Advanced CFD (ACFD)
- Advanced Thermodynamics
- Continuum Mechanics
- Gas Dynamics
- Computational Mechanics Lab

CERTIFICATIONS

- CATIA & ANSYS** Hyderabad
Central Institute of Tool Design (MSME) Dec. 2020 – Jan. 2021